



► ILO Brief

July 2026

Generative AI and labour markets in ASEAN: Significant exposure, limited disruption, uneven preparedness

Key points

- **Exposure is significant, but high exposure remains limited.** As AI technologies continue to advance and diffuse across Association of Southeast Asian Nations (ASEAN) economies, nearly 80 million workers in ASEAN are employed in occupations with some degree of potential exposure to generative artificial intelligence (GenAI), equivalent to 22.9 per cent of total employment. However, only 3.3 per cent of employment is concentrated in occupations with the highest levels of exposure. GenAI exposure measures the technical feasibility of automating or assisting with tasks within occupations.
- **Women are overrepresented in occupations with the highest levels of GenAI exposure.** Across ASEAN, the share of women employed in these occupations is nearly twice that of men, reflecting their concentration in clerical, administrative and selected professional occupations.
- **GenAI adoption remains concentrated in digitally intensive occupations.** Workplace use of GenAI is highest in technology-related and knowledge-intensive occupations, while remaining comparatively limited in many clerical and administrative occupations, despite their high exposure.
- **Widespread labour market disruption is not yet visible.** Employment in highly exposed occupations has continued to expand across ASEAN, with no evidence so far of large-scale job losses associated with the emergence of GenAI.
- **Employment growth in some highly exposed occupations has been weaker among young workers than among adults.** Several indicators point to slower growth or declining employment among young workers in highly exposed occupations and selected entry-level jobs. However, these trends have not been accompanied by marked increases in youth unemployment or the share of youth not in education, employment or training (NEET).
- **Future outcomes will depend on policy choices today.** The extent to which GenAI contributes to productivity gains, quality jobs and inclusive and equitable growth will depend on countries' level of preparedness, which differs considerably across ASEAN. Enhanced preparedness will require investments in digital infrastructure, skills development, enterprise capabilities and social protection. Integrated governance frameworks and coordination mechanisms, informed by evidence and underpinned by social dialogue, are essential to translate assessments of GenAI exposure and adoption into effective policy action, ensuring that digital economy and sectoral policies incorporate decent work dimensions.

► Introduction

Generative artificial intelligence (GenAI) is rapidly expanding the range of tasks that can be automated or augmented, including many cognitive and non-routine activities previously considered difficult to perform with digital technologies. As GenAI capabilities and adoption continue to advance, strengthening policy responses and governance frameworks is becoming increasingly important to address the implications for employment, job quality, skills demand and labour market inequalities. While much of the global debate has focused on the potential risks of job displacement, the ultimate effects of GenAI are likely to depend on how technologies are adopted, how work is reorganized, and how workers, enterprises and institutions adapt to change.

The implications of GenAI for the Association of Southeast Asian Nations (ASEAN) region are likely to be diverse and uneven. ASEAN economies differ considerably in their levels of economic development, demographic profiles and employment structures, as well as digital infrastructure, technological capabilities and labour market institutions. At the same time, AI adoption is evolving alongside broader digital transformations, economic structural change and demographic shifts that are shaping labour markets.

In this context, occupational exposure to GenAI does not automatically translate into labour market change. Exposure identifies where AI could potentially affect work tasks and reflects the technical feasibility of task automation or augmentation. Actual outcomes will depend on the pace and breadth of adoption, including the uptake across sectors and enterprises of different sizes, workforce capabilities and the broader policy and institutional environment.

Building on earlier research,¹ this brief examines the implications of GenAI for jobs and labour markets across ASEAN through three complementary perspectives. First, it assesses the region's potential for labour market transformation associated with GenAI by examining both occupational exposure and emerging patterns of AI adoption. Together, these indicators provide insights into where labour market changes may be most likely to emerge. Second, the brief examines whether impacts are already becoming visible by analysing recent employment trends in occupations with high levels of GenAI exposure. Finally, it considers the preparedness of ASEAN economies to harness the opportunities associated with AI while ensuring labour market inclusion and resilience. This includes an assessment of digital infrastructure, skills, governance frameworks and broader institutional conditions that will shape how GenAI affects workers, enterprises and labour markets in the years ahead.

The purpose of this brief is to provide an evidence base for policy discussions and future research on the implications of GenAI for jobs and labour markets in ASEAN. The evidence presented in this brief suggests that the potential for labour market transformation is significant, but that widespread disruption is not yet visible. While AI adoption is increasing, it remains uneven across countries, sectors and occupations. Future outcomes will therefore depend not only on technological progress, but also on policy choices today. The extent to which ASEAN countries are prepared to manage this transition, will help determine whether AI contributes to productivity gains, quality jobs and inclusive and equitable growth.

► GenAI and the potential for labour market transformation in ASEAN

This section examines the potential for AI-driven labour market transformation in ASEAN by considering both occupational exposure and emerging patterns of GenAI adoption. Occupational exposure provides an indication of

where GenAI could potentially affect work tasks, while evidence on adoption shows the extent to which workers and enterprises are already integrating GenAI into their activities. Together, these indicators help identify where

¹ ILO, "Generative AI and Jobs in the Philippines: Labour Market Exposure and Policy Implications", ILO Policy Brief, 2026; ILO, "Generative AI and Jobs in Viet Nam: Labour Market Exposure and Policy Considerations", ILO Brief, 2026.

labour market changes may be most likely to emerge and provide important context for interpreting subsequent labour market developments.

The occupational exposure analysis draws on the methodology developed by Gmyrek et al. (2025),² which assesses the extent to which tasks performed within occupations could technically be automated or assisted by GenAI. Occupations are classified into six categories, dependent on the level of GenAI exposure: (i) no exposure, (ii) minimum exposure, (iii) low exposure, (iv) moderate exposure, (v) significant exposure and (vi) high exposure (see [Annex 1](#) for a detailed description of the six exposure categories). Higher exposure categories indicate that a larger share of tasks within an occupation could potentially be automated or assisted by GenAI.

It is important to note that exposure reflects technical feasibility of automation or augmentation, rather than actual labour market outcomes, which depend on technology adoption, workplace organization, skills and the broader policy environment.

Nearly one in four workers are potentially exposed to GenAI, but high exposure remains limited

According to ILO estimates for 2025, 22.9 per cent of total employment in ASEAN is in occupations with more than a minimal degree of potential exposure to generative AI, equivalent to nearly 80 million workers. Most exposed workers are concentrated in occupations with relatively low or moderate levels of exposure. Only 3.3 per cent of total employment, corresponding to 11.7 million workers, were employed in occupations classified within the highest exposure category (G4) (figure 1).³ At the same time, around 67 per cent of employment remained concentrated in occupations with no identified exposure to GenAI.

These patterns suggest that, while GenAI has the potential to affect a sizeable share of workers across ASEAN, occupations with the highest levels of exposure still account for a relatively small share of total employment.

At the country-level, across the nine ASEAN countries with available data, Singapore records the highest share of workers with more than a minimal degree of GenAI exposure, accounting for 42.2 per cent of total employment. This may, in part, reflect the country's highly knowledge-intensive occupational structure, where professionals, managers, executives and technicians account for around 64 per cent of employed residents, concentrated in financial and insurance services, information and communications, and professional services.⁴ This is followed by the Philippines (28.1 per cent), which reflects, in part, the country's relatively service-oriented economy and the prominence of the information technology and business process management sector.⁵ Indonesia (21.7 per cent), Viet Nam (20.8 per cent) and Thailand (20.6 per cent) also record significant levels of exposure. The share of employment in the highest exposure category (G4) ranges from 0.3 per cent in Timor-Leste to 6.3 per cent in Singapore.⁶

Differences in the sectoral structure of ASEAN economies are likely to play an important role in explaining cross-country variation in GenAI exposure. Existing evidence suggests that exposure tends to be concentrated in service-sector activities that rely heavily on information processing, administrative work and knowledge-intensive tasks, including finance and insurance, information and communications, professional and business services, public administration and education.⁷ Some of these sectors are also often associated with relatively higher earnings, although patterns vary across countries (box 1). As a result, exposure levels depend not only on occupational characteristics, but also on the relative importance of these sectors within individual economies.

² See: Paweł Gmyrek et al., "Generative AI and Jobs: A Refined Global Index of Occupational Exposure", ILO Working Paper No. 140, May 2025. Due to data constraints, the methodology applied in this brief does not take into account disparities in digital infrastructure across countries, which may limit actual GenAI exposure and needs to be taken into account when interpreting the findings. See: Paweł Gmyrek, Mariana Viollaz and Hernan Winkler, "Disruption without Dividend? How the Digital Divide and Task Differences Split GenAI's Global Impact", ILO Working Paper No. 166, March 2026.

³ See [Annex 2](#) for a list of occupations that form part of the Gradient 4 classification.

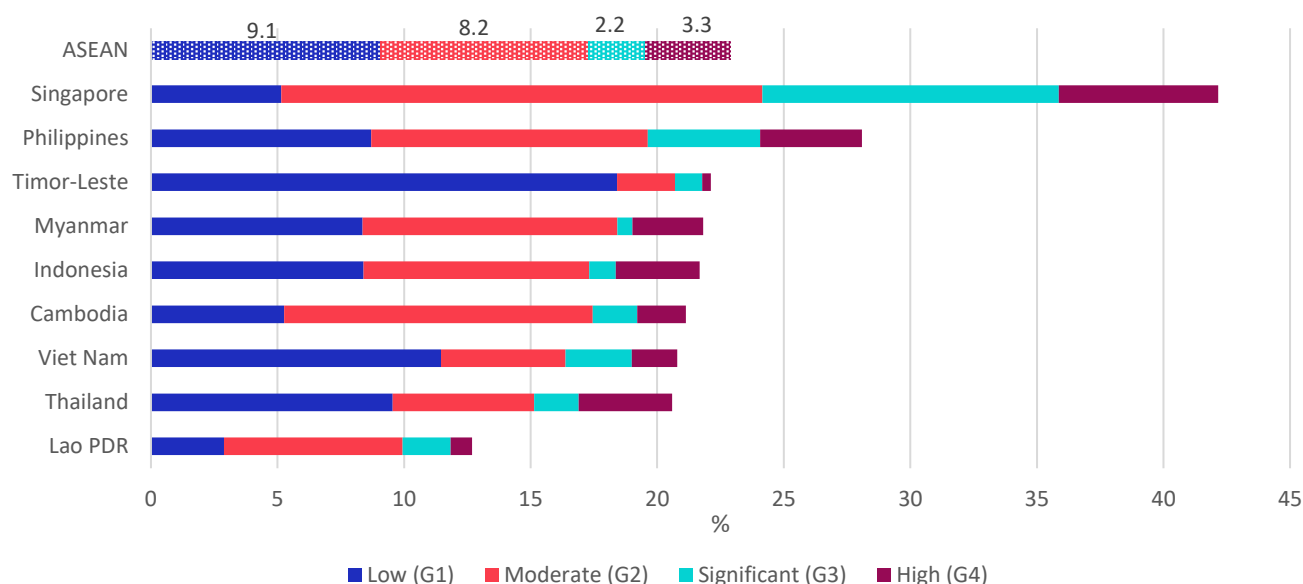
⁴ See: Singapore, Ministry of Manpower, *Labour Force in Singapore Advance Release 2025*, 2025.

⁵ See: Kristhel Anne M. Caynila and Genna Paola S. Centeno, "Adopting Generative Artificial Intelligence: Opportunities and Challenges in the Philippine IT-BPM industry", *Central Bank of the Philippines Economic Newsletter* No. 25-01, July 2025.

⁶ Data points for Malaysia are not shown in the figure, as micro labour force survey data are not available to the ILO. For some evidence on GenAI exposure in Malaysia, see Calvin Cheng et al., *Novel AI Technologies and the Future of Work in Malaysia* (World Bank, 2025).

⁷ Flavio Calvino, Daniel Haerle and Sarah Liu, "Is Generative AI a General Purpose Technology?", OECD Artificial Intelligence Paper No. 40, June 2025; Flavio Calvino et al., "A Sectoral Taxonomy of AI Intensity". OECD Artificial Intelligence Paper No. 30, December 2024.

► **Figure 1. Employment by level of Gen AI exposure, latest available year (% of total employment)**

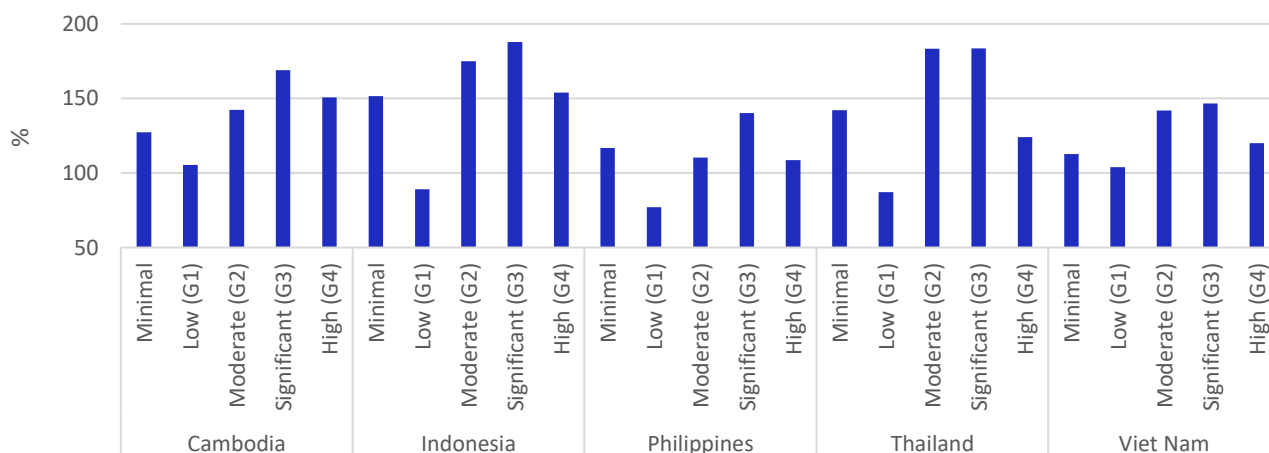


Notes: Figures refer to ASEAN (2025), Cambodia (2023), Indonesia (2023), Lao People's Democratic Republic (2017), Myanmar (2020), Philippines (2023), Singapore (2023), Thailand (2024), Timor-Leste (2022) and Viet Nam (2024). See [Annexes 1–3](#) for more information on GenAI exposure categories and estimation methodology used to produce estimates for ASEAN. Source: ILO estimates based on ILO Microdata Repository and Gmyrek et al. 2025. For Singapore, the data point is based on Gmyrek, Viollaz and Winkler 2026.

► **Box 1. Higher earnings in occupations with significant GenAI exposure**

An analysis of GenAI exposure and wages reveals a consistent pattern, where occupations with significant GenAI exposure (G3) were also the highest paid (figure B1).¹ Average hourly earnings tend to increase with higher occupational exposure to GenAI, though the strength of this relationship varies by country.

► **Figure B1. Average hourly earnings of wage employees in selected countries by level of GenAI exposure, latest available year (% of earnings of non-exposed)**



Notes: Earnings of non-exposed employees are set to 100 per cent. Figures refer to Cambodia (2023), Indonesia (2023), Philippines (2023), Thailand (2024) and Viet Nam (2024). See [Annexes 1 and 2](#) for more information on GenAI exposure categories. Source: ILO estimates based on ILO Microdata Repository and Gmyrek et al. 2025.

In Cambodia, Indonesia, the Philippines and Viet Nam, wage workers with significant GenAI exposure earned notably more than those who faced minimal or no exposure, suggesting that GenAI-exposed roles are associated with higher productivity or skill premiums. By comparison, Thailand reveals a similar trend but the association between greater GenAI exposure and higher earnings is less clear especially between wage earners in the moderate exposure (G2) and significant exposure (G3) categories. Overall, these patterns point to the need for the right policy mix to ensure that the gains from GenAI are not concentrated among already high-paid workers but are distributed in an inclusive and equitable manner.

¹ Gradient 3 occupations comprise of relatively high-paying professions, including accountants (2411), financial and investment advisers (2412), software developers (2512), application programmers (2514), database designers and administrators (2521), systems administrators (2522) and economists (2631), among others. ISCO codes are reported in parentheses.

Women are more likely than men to work in highly exposed occupations

Women are substantially more likely than men to work in occupations with the highest levels of GenAI exposure. Across ASEAN, 4.8 per cent of women were employed in occupations with high GenAI exposure (G4) in 2025, compared with 2.3 per cent of men (figure 2a). The gender gap is most pronounced in Thailand and the Philippines, where women are around three to four times more likely than men to work in highly exposed occupations. Similar, though smaller, gaps are observed across most other ASEAN countries.

These patterns reflect the concentration of women in clerical, administrative and some professional occupations that are more exposed to GenAI, while men remain more concentrated in manual and operational occupations with lower exposure levels. As a result, GenAI may affect women and men differently, with potential implications for gender inequalities and labour market inclusion. At the same time, women remain underrepresented in many STEM⁸ and AI-related occupations and the design and governance of AI systems, which may limit their access to some of the emerging opportunities associated with the development and deployment of AI technologies.⁹

Youth and adult exposure to GenAI are broadly comparable

Youth (ages 15–24) and adult (ages 25+) workers exhibit broadly similar levels of exposure to GenAI. Across ASEAN, the share of employment in the highest exposure category (G4) is comparable for young workers and adult workers (figure 2b).

In several countries, however, young workers are slightly more likely than adults to be employed in highly exposed occupations. The difference is most pronounced in Indonesia and the Philippines, where 4.0 and 4.6 per cent of youth employment, respectively, is concentrated in the high exposure category (G4), compared with 3.2 and 3.9 per cent among adults. Similar, though generally smaller, differences are observed in Thailand.

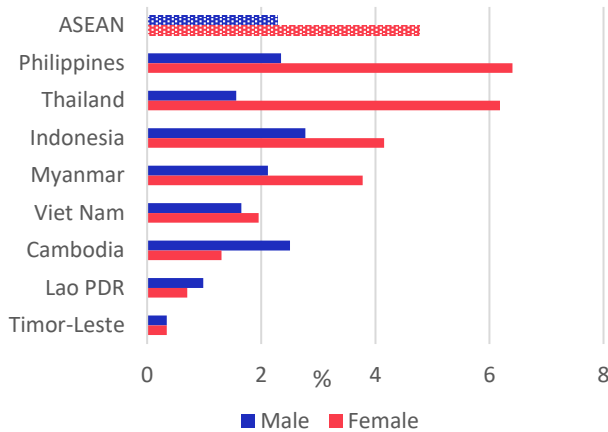
The patterns in these countries reflect the concentration of younger workers in clerical, administrative and other entry-level occupations that contain a relatively large share of tasks that can potentially be automated or transformed through GenAI technologies. While age-related differences in exposure remain modest for the region overall, the findings highlight the importance in some countries of supporting school-to-work transitions and ensuring that education and training systems equip young people with adaptable skills that remain relevant as AI continues to reshape work and occupations.

⁸ STEM refers to science, technology, engineering and mathematics.

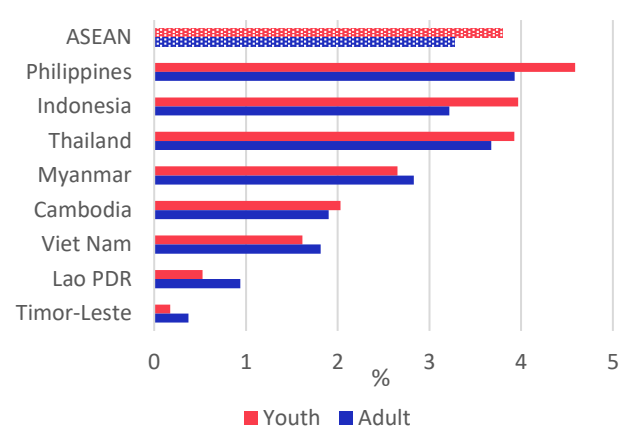
⁹ For more a more detailed discussion and evidence, see ILO, “Gen AI, Occupational Segregation and Gender Equality in the World of Work”, ILO Research Brief, March 2026.

► **Figure 2. Employment with high Gen AI exposure (G4), latest available year (% of total employment)**

a. By sex



b. By age group



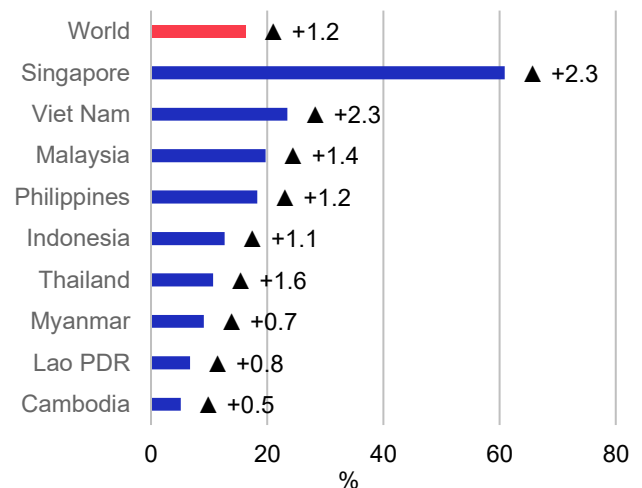
Notes: "Youth" are those aged 15–24 years, and "adults" are those aged 25+ years. Figures refer to ASEAN (2025), Cambodia (2023), Indonesia (2023), Lao People's Democratic Republic (2017), Myanmar (2020), Philippines (2023), Thailand (2024), Timor-Leste (2022) and Viet Nam (2024). See [Annexes 1–3](#) for more information on GenAI exposure categories and estimation methodology used to produce estimates for ASEAN. Source: ILO estimates based on ILO Microdata Repository and Gmyrek et al. 2025.

AI adoption is rising, but remains uneven across ASEAN

Occupational exposure to GenAI does not by itself determine labour market outcomes. Instead, it signals where jobs and tasks could be affected, with actual workplace transformation dependent on the decisions of workers and employers to adopt and integrate AI technologies.

In practice, AI adoption tends to diffuse gradually rather than uniformly across the economy.¹⁰ Adoption often begins in technologically intensive occupations and sectors, where digital capabilities and complementary skills are already well established, before spreading more broadly over time.¹¹ Firm size also appears to matter, with large firms often leading AI adoption, while adoption among smaller firms remains more limited (see box 2 for evidence from Singapore). Larger firms are typically better positioned to adopt AI because they benefit from stronger data infrastructures, clearer governance structures and greater access to specialized technical talent.¹²

► **Figure 3. Share of people who have used a GenAI product, selected economies and worldwide, 2nd half 2025 (% change from 1st half 2025)**



Notes: AI diffusion defined as the share of people worldwide who have used a GenAI product during the reported period. It does not reflect GenAI adoption at the enterprise level or in the workplace only. Source: Microsoft AI Economy Institute, *Global AI Adoption in 2025: A Widening Digital Divide*, 2026.

¹⁰ The gradual diffusion of new technologies is consistent with broader evidence that productivity gains often materialize only after complementary investments and organizational adjustments have been made. See Erik Brynjolfsson, Daniel Rock and Chad Syverson, "The Productivity J-Curve: How Intangibles Complement General Purpose Technologies", *American Economic Journal: Macroeconomics* 13, No. 1 (2021): 333–372.

¹¹ Gabriel Demombynes, Jörg Langbein and Michael Weber, "From Predictions to Practice: What AI Usage Data Reveals About the Future of Work", *World Bank Protect and Invest in People* (blog), 19 February 2026.

¹² See also ILO, *Adoption of Artificial Intelligence and Productivity Growth in Kuwait's Private Sector: Enterprise Perspectives and Evidence*, 2026.

In ASEAN economies, recent data point to uneven but rising use of GenAI. The share of people using a GenAI product increased across all countries between the first and second half of 2025, with particularly strong uptake in Singapore and Viet Nam (figure 3). While usage in several ASEAN countries remains below the global average, the upward trend suggests a gradual diffusion of GenAI technologies across the region, though this reflects general uptake and not solely adoption by enterprises or workplaces.

How GenAI is used in practice, particularly for work-related tasks, provides a more nuanced view of its diffusion. Recent Anthropic research suggests that observed GenAI exposure remains below what is theoretically feasible.¹³ Within ASEAN, usage of Anthropic's Claude.ai for work purposes as a share of overall use varies widely, ranging from 29.4 per cent in Indonesia to 52.0 per cent in Viet Nam.¹⁴

GenAI adoption remains concentrated in technology intensive occupations

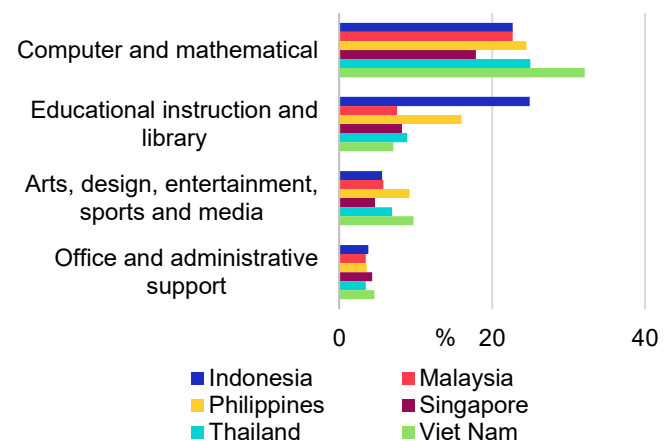
While the use of Claude.ai is not necessarily representative of overall GenAI use, usage data reveal some notable patterns, as GenAI use of Claude.ai varies considerably across occupational groups (figure 4). Across ASEAN economies with available Claude.ai data, GenAI use is most prevalent in computer and mathematical occupations, such as software developers, followed by education-related roles, including tutors, reflecting its concentration in more technology-intensive and knowledge-based activities. Notably, observed use remains relatively limited in office and administrative support occupations, despite these roles featuring prominently among highly exposed occupational groups (Gmyrek et al., 2025) (see [Annex 2](#) for a detailed list of occupations with high GenAI exposure (G4)).

Realizing the benefits of GenAI for productivity and decent work will require continued investments in complementary skills,¹⁵ firm capabilities, institutional

readiness and enabling infrastructure. Firm-level evidence on AI adoption remains limited across ASEAN. However, Singapore's recent enterprise survey data provide valuable insights into emerging patterns of AI adoption among firms (box 2).

Taken together, the evidence suggests that the potential for GenAI-driven labour market transformation in ASEAN is significant, but that adoption remains at a relatively early and uneven stage. While a sizeable share of workers are employed in occupations with some degree of exposure to GenAI, actual use of AI technologies remains concentrated in selected countries, sectors and occupations. As a result, occupational exposure should not be interpreted as a measure of labour market impact. Rather, exposure and adoption patterns provide an indication of where labour market changes may be most likely to emerge as AI diffusion continues across the region. The following section examines whether such changes are already becoming visible in labour market outcomes across ASEAN economies.

► **Figure 4. Share of Anthropic's GenAI (Claude.ai) conversations by top four job groups and country, February 2026 (%)**



Notes: Percentages represent the share of Anthropic's Claude.ai conversations associated with each task or group of tasks for the period 5 February to 12 February 2026. Data is categorized using O*NET-SOC codes and shows what people do with Claude, not their job titles. Data includes ASEAN economies with more than 3,000 observations during the reporting period. Source: Anthropic Economic Index available at <https://www.anthropic.com/economic-index#country-usage>.¹⁶

¹³ Maxim Massenkoff and Peter McCrory, "Labor Market Impacts of AI: A New Measure and Early Evidence", *Anthropic Research* (blog), 5 March 2026.

¹⁴ Anthropic Economic Index available at: <https://www.anthropic.com/economic-index#country-usage>. Usage data from Anthropic's Claude.ai for the period 5 February to 12 February 2026 available at: <https://huggingface.co/datasets/Anthropic/EconomicIndex> (accessed 5 May 2026). The findings based on data from a single GenAI platform (Anthropic's Claude) are not representative of overall generative AI usage patterns.

¹⁵ See also ILO, *Lifelong Learning and Skills for the Future*, World of Work Series, 2026.

¹⁶ The analysis in this brief focuses on GenAI and does not encompass the wider spectrum of advanced technologies already adopted across sectors in ASEAN, such as industrial robotics and advanced process automation.

► **Box 2. Singapore: AI adoption is still at an early stage and driven by large firms**

In Singapore, an establishment-based survey in 2026 by the Ministry of Manpower shows that AI adoption at the firm level remains at an early stage and is concentrated among large firms. A vast majority (71.5 per cent) of the surveyed firms reported that they had not begun AI adoption in their operations; only 3.8 per cent of enterprises had integrated AI in their core business processes. Larger firms with more than 500 employees reported considerably higher rates of adoption (76.4 per cent) and more extensive integration than smaller firms; for example, only 23.9 per cent of firms with less than 25 employees reported to have adopted AI. AI adoption was most prevalent in knowledge-intensive sectors such as information and communication (74.1 per cent), professional services (57.5 per cent), and financial and insurance services (56.4 per cent), compared with much lower levels in administrative and support services (21.1 per cent). Most firms (70.7 per cent) using AI reported increased worker productivity. Reported consequences of AI adoption ranged from redesign of job roles and creation of new AI-related jobs, to redeployment of workers to new roles and changes in hiring activity, indicating that AI is primarily transforming tasks rather than replacing occupations.

Note: Based on establishment-based survey covering 2,560 private sector establishments employing 486,600 workers, with fieldwork conducted between January and March 2026. Source: Singapore, MOM, Manpower Research and Statistics Department, “[Adoption of Artificial Intelligence Among Firms](#)”, April 2026.

► **Are labour market impacts of GenAI emerging in ASEAN?**

This section examines employment trends in ASEAN across occupations with differing levels of GenAI exposure. While the current wave of GenAI is widely associated with the public release of ChatGPT in late 2022, the occupations potentially exposed to these technologies have long been embedded within ASEAN labour markets. Examining how employment in these occupations has evolved over time, including before and after 2022, provides insights into whether employment shifts are emerging in occupations that are more exposed to GenAI.

While such analysis cannot establish causality – as employment trends are shaped by a range of economic, demographic and technological factors – it can help identify patterns that may warrant closer monitoring as AI adoption expands.

Employment exposed to GenAI is increasing in ASEAN economies

In ASEAN, employment in occupations with at least minimal potential exposure to GenAI increased steadily between 2017 and 2025 (figure 5a). Employment in occupations with low, moderate, significant or high GenAI exposure rose from around 66 million workers in 2017

(corresponding to 20.9 per cent of total employment) to 74 million in 2022 (22.2 per cent) and 80 million in 2025 (22.9 per cent).

This indicates that these occupations were already expanding prior to the emergence of GenAI, with no clear evidence of subsequent employment declines. At the same time, this pattern should not be interpreted as evidence that GenAI itself is driving employment growth. Rather, it is consistent with broader trends of structural transformation, rising demand for service-sector and knowledge-intensive work, and continued population and labour force growth across ASEAN economies.

Faster growth in higher exposure occupations reflects both digitalization and economic development

Occupations with more than minimal GenAI exposure recorded relatively strong employment growth both before and after 2022. In the period 2017–2022, employment growth was highest among occupations in the highest exposure category (G4), followed by occupations with significant (G3) and moderate (G2)

exposure. A similar pattern is observed during 2022–2025, although growth in high exposure occupations (G4) moderated, while occupations with significant (G3) exposure recorded somewhat stronger growth. By contrast, occupations with no exposure to GenAI continued to exhibit comparatively modest employment growth (figure 5b). Consistent with these trends, also the share of employment accounted for by occupations in the highest exposure category (G4) continued to increase after 2022 (figure 6a).

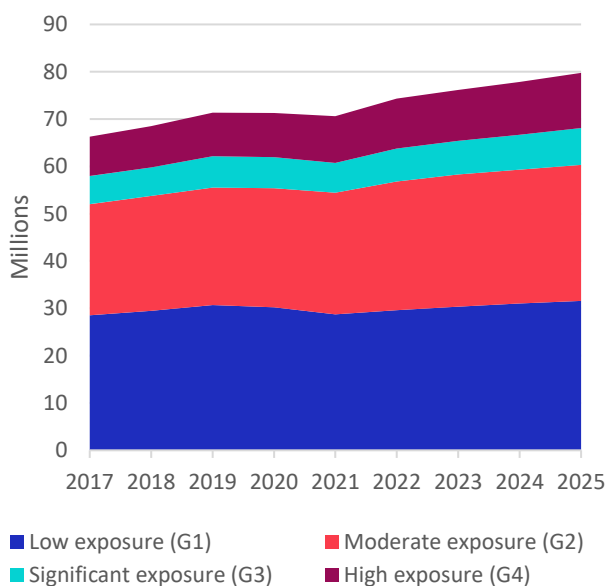
The continued expansion of these occupations suggests that there is currently little evidence of widespread employment decline in highly exposed occupations. Rather, the findings point to the growing labour market importance of occupations in which GenAI may increasingly reshape tasks and work processes. Exposure

reflects the technical feasibility of automating or augmenting selected tasks within occupations, many of which continue to rely on human judgement, interpersonal interaction and other complementary skills.

The absence of widespread labour market disruption is also consistent with emerging international evidence suggesting that, while GenAI can generate meaningful productivity gains at the task level, these gains have not yet translated into measurable changes in aggregate productivity and employment.¹⁷ This may also reflect the fact that AI adoption still remains at an early stage. AI may become a labour-saving technology and lead to more pronounced employment effects only after full integration into production processes at a large scale, which may take time to materialize.¹⁸

► **Figure 5. Employment in ASEAN by level of GenAI exposure**

a. Total employment, 2017–2025 (millions)



b. Average annual employment growth, 2017–2022 and 2022–2025 (%)



Note: See [Annexes 1–3](#) for more information on GenAI exposure categories and estimation methodology. Source: ILO estimates based on ILO Microdata Repository and Gmyrek et al. 2025.

¹⁷ See also Anders Humlum and Emilie Vestergaard, “Large Language Models, Small Labor Market Effects”, Becker Friedman Institute for Economics Working Paper No. 2025-56, April 2025; Brynjolfsson, Rock and Syverson 2021.

¹⁸ Ivan Yotzov et al., “Firm Data on AI”, NBER Working Paper No. 34836, February 2026.

Employment trends in highly exposed occupations differ across countries and age groups

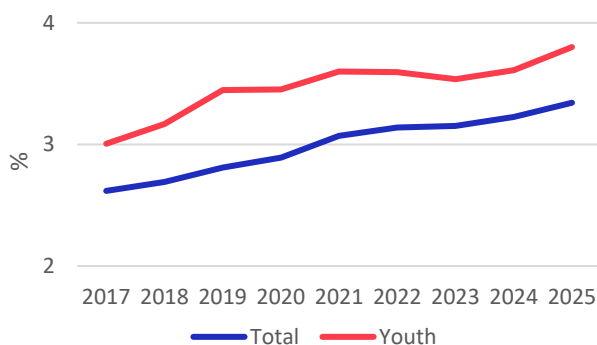
Beneath the regional trends, country-level developments differ. Thailand exhibited little overall change, while the Philippines recorded a noticeable decline in the share of employment in highly exposed occupations over the course of 2023. By contrast, Viet Nam experienced a gradual increase from a lower initial level. Taken together, these patterns suggest that labour market developments in highly exposed occupations continue to be shaped primarily by country-specific economic and structural factors rather than a common regional response to GenAI.

As well, some distinct differences emerge for younger workers. In both the Philippines and Thailand, youth employment in highly exposed occupations declined markedly relative to total youth employment, whereas youth and overall employment trends remained broadly aligned in Viet Nam. During the same period, available evidence does not point to a corresponding increase in youth unemployment or the rate of youth not in employment, education or training (NEET).¹⁹

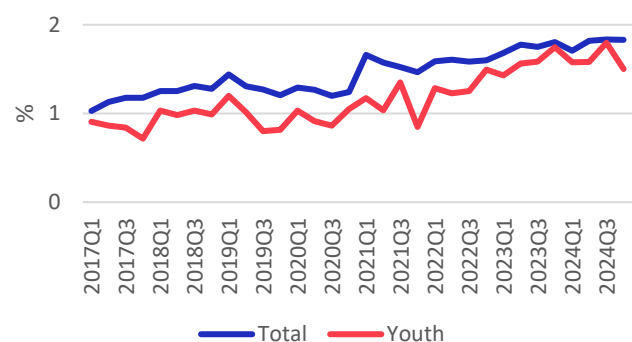
This suggests that recent developments may reflect changes in hiring patterns, different education choices, occupational reallocation or evolving entry pathways into highly exposed occupations, rather than indicating a broad weakening of labour market outcomes for young people associated with GenAI (see also box 3).

► **Figure 6. Share of employment with high GenAI exposure (G4), 2017–2024 (%)**

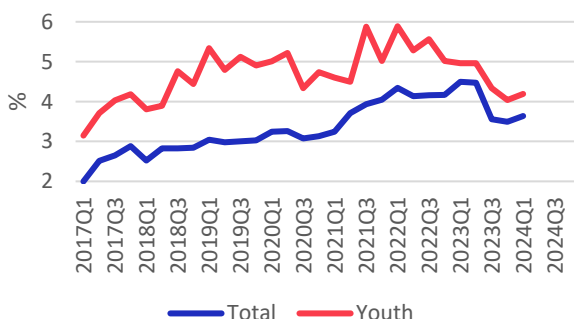
a. ASEAN



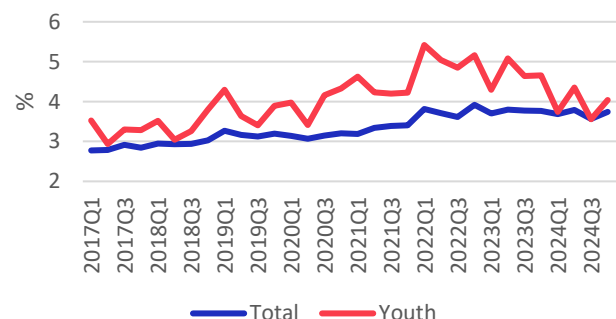
b. Viet Nam



c. Philippines



d. Thailand



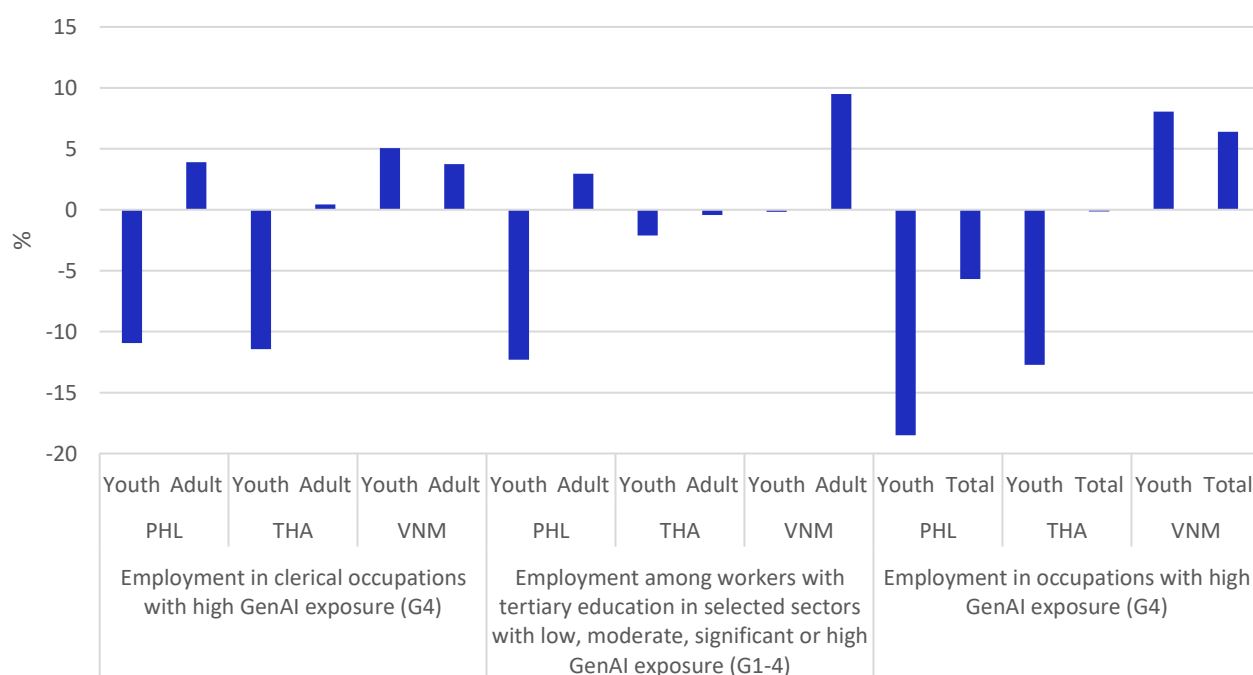
Notes: Youth aged 15–24 years. Data for Philippines and Viet Nam include a series break in Q1-2021 and Q1-2019, respectively. The data point for ASEAN is based on an ILO estimate; see [Annexes 1-3](#) for more information on GenAI exposure categories and estimation methodology. Source: ILO estimates based on ILO Microdata Repository and Gmyrek et al. 2025.

¹⁹ In Philippines, the youth unemployment rate remained constant at 6.9 per cent between 2022 and 2023, while the NEET rate slightly decreased from 13 to 12.4 per cent during the same time period. In Thailand, the youth unemployment rate declined from 5.1 per cent in 2022 to 4.6 per cent in 2024; during the same period, the NEET rate declined from 13.3 to 12.8 per cent. See ILOSTAT database for data based on national labour force surveys.

► **Box 3. Young workers in a changing world of work: Early signs of labour market change**

While the evidence remains preliminary and cannot be attributed directly to GenAI adoption, a consistent pattern emerges across several indicators. Recent labour market developments in highly exposed occupations suggest that employment outcomes for young workers may be evolving differently from those of adults (figure B2). This is consistent with a broader discussion in the literature that young workers may act as “canaries in the coal mine”, with changes in entry-level employment and school-to-work transitions potentially becoming visible before broader labour market effects emerge.¹

Across the Philippines and Thailand, youth employment in highly exposed occupations declined more rapidly than overall employment, while Viet Nam continued to record positive employment growth. Similar patterns are observed in highly exposed clerical occupations, where employment among young workers fell sharply in the Philippines and Thailand. Employment outcomes for tertiary-educated youth in selected GenAI-exposed sectors were also generally weaker than for adults, suggesting that higher educational attainment alone may not fully offset challenges associated with changing skill demands and labour market entry. Taken together, these findings do not provide evidence of widespread labour market disruption, but suggest that school-to-work transitions and entry-level employment opportunities warrant continued monitoring as AI adoption expands across ASEAN economies.

► **Figure B2. Average annual change in employment among workers in specific groups, 2022–2024 (%)**

¹ See Erik Brynjolfsson, Bharat Chandar and Ruyu Chen, “Canaries in the Coal Mine? Six Facts about the Recent Employment Effects of Artificial Intelligence”, Stanford Digital Economy Lab Working Paper, 2025.

Note: PHL = Philippines, THA = Thailand, VNM = Viet Nam. Youth aged 15–24 years; adults aged 25+ years. Figures correspond to the Philippines (Q1), Thailand (annual) and Viet Nam (annual). Selected sectors include the following ISIC Rev. 4 sections: G – Wholesale and retail trade; J – Information and communication; K – Financial and insurance activities; M – Professional, scientific and technical activities; and N – Administrative and support service activities. Clerical occupations include: 4110 – General office clerks, 4131 – Typists and word processing operators, 4132 – Data entry clerks, 4311 – Accounting and bookkeeping clerks, 4312 – Statistical, finance and insurance clerks, 4313 – Payroll clerks, 4416 – Personnel clerks and 4419 – Clerical support workers not elsewhere classified. Source: ILO estimates based on ILO Microdata Repository and Gmyrek et al. 2025.

► Is ASEAN prepared for an AI-driven future of work?

Overall, the available evidence does not point to widespread labour market disruption associated with GenAI in ASEAN economies. As the diffusion of GenAI remains at a relatively early stage, the extent to which future labour market outcomes translate into productivity gains, job creation or adjustment pressures will depend not only on technological developments, but also on the preparedness and resilience of workers, firms and institutions to adapt to them.

Harnessing the benefits of GenAI requires more than access to technology. Productivity gains depend on complementary investments in digital infrastructure, connectivity, computing capacity, enterprise capabilities and workforce skills and on addressing systemic and structural barriers faced by women and other marginalized groups. Equally important are governance frameworks, social protection systems, labour market institutions and social dialogue mechanisms that can support workplace transitions, promote responsible AI use and help ensure that productivity gains are equitably shared. Countries that combine strong digital foundations with investments in human capital and institutional capacity are likely to be better positioned to translate AI adoption into productivity growth, quality job creation and inclusive development.

The diffusion of GenAI across ASEAN is taking place against a backdrop of uneven digital skills readiness. Available ICT skills data reveal substantial differences across countries.²⁰ Singapore and Malaysia report relatively high levels of advanced digital skills across several domains, while other countries continue to face significant gaps in areas such as problem-solving, digital content creation and communication skills. At the same time, preparedness requires more than technical AI skills alone, and is likely to depend on a broader combination of digital, cognitive and socio-emotional skills that enable workers and firms to adapt to changing technologies and work processes.

Governance frameworks are expanding, but remain focused on skills and adoption

At the regional level, ASEAN has adopted a largely soft-law and coordination-oriented approach to AI governance. Key frameworks – including the [ASEAN Digital Masterplan 2025](#), the [ASEAN Responsible AI Roadmap \(2025–2030\)](#), and the [ASEAN Guides on AI Governance and Ethics](#) – promote a human-centred and innovation-friendly approach, while leaving implementation and regulation largely to national governments.

These initiatives place strong emphasis on capacity-building, skills development and responsible innovation. Labour and social dimensions – including digital inclusion, gender responsiveness and equity in AI systems – are recognized, including potential risks relating to employment disruption and social protection, although governance remains focused primarily on guidance, cooperation and multi-stakeholder engagement rather than binding labour standards or formalized tripartite mechanisms.

National AI strategies broadly reflect this approach. Across most ASEAN economies, policy frameworks prioritize workforce adaptation, skills development, AI adoption and innovation (see [Annex 4](#) for an overview of existing policies and strategies). Malaysia has integrated labour market, skills and inclusion considerations particularly explicitly through its National AI Roadmap and AI Governance and Ethics Guidelines. Other countries, including Indonesia, the Philippines, Singapore, Thailand and Viet Nam have developed increasingly comprehensive strategies combining talent development, sectoral adoption and governance objectives. Notwithstanding the integration of GenAI dimensions within broader planning, or specific educational or employment frameworks, when considering AI policies themselves, the sophistication of national frameworks varies, and the common denominator seems to be for most ASEAN economies to strengthen AI capabilities while preparing workers and enterprises for technological change, with lesser consideration for other work-related impacts. A key

²⁰ See the ITU Data Hub: <https://datahub.itu.int/>. Data are available for Brunei Darussalam, Indonesia, Malaysia, the Philippines and Singapore.

challenge is therefore to ensure that AI governance does not remain confined to technology, innovation and skills agendas alone. Employment, social protection, job quality, enterprise development and social dialogue considerations need to be more systematically reflected in digital economy and sectoral policy frameworks.

AI readiness varies significantly across ASEAN economies

Despite growing policy activity, readiness to adopt and benefit from AI remains uneven across ASEAN. Indicators of AI preparedness reveal substantial differences in digital infrastructure, human capital, innovation ecosystems, governance frameworks and access to finance (see [Annex 5](#)). For reference, the International Monetary Fund (IMF) AI Preparedness Index assigns the ASEAN-5 countries²¹ an average score of 0.60, compared with 0.68 for advanced economies, although differences within the region are considerable.

Importantly, labour market exposure and AI preparedness do not necessarily always move together (box 4). Countries with similar levels of exposure to GenAI often display markedly different levels of preparedness, suggesting that the capacity to manage AI-driven transformation varies considerably across ASEAN. Singapore stands out as a notable exception, combining both high exposure and high preparedness. These differences across the region suggest that future labour market outcomes will depend not only on where workers are exposed to AI, but also on the capabilities available to support resilience, innovation and productivity growth.

A three-tier landscape of AI readiness is emerging

Taken together, the evidence points to a three-tier landscape of AI readiness across ASEAN.

Singapore stands out as a globally competitive AI ecosystem, combining advanced digital infrastructure, strong talent availability and a whole-of-government approach to implementation. While challenges remain²² in

strengthening frontier innovation, Singapore is well positioned to harness the opportunities associated with AI adoption.

A second group – including Malaysia and Thailand, followed by Brunei Darussalam, Indonesia, the Philippines and Viet Nam – has established many of the foundations needed for AI adoption but continues to face important gaps in areas such as advanced skills, research capacity, computing infrastructure, innovation ecosystems and access to finance.²³ The nature and severity of these constraints vary across countries, but they are likely to influence both the pace of AI adoption and the ability to translate technological advances into productivity gains.

Cambodia, the Lao People's Democratic Republic, Myanmar and Timor-Leste generally exhibit lower levels of readiness, reflecting more limited digital infrastructure, institutional capacity and technological capabilities. While these countries may initially face lower levels of exposure to GenAI, they also risk capturing fewer of the potential benefits associated with AI-driven productivity growth.

AI readiness should not be analysed in isolation. As several ASEAN economies have built growth strategies around labour-intensive manufacturing, export-oriented services or business process outsourcing, the ability to adopt and effectively use GenAI may increasingly influence the competitiveness of these sectors. More broadly, GenAI may affect the long-term viability of these development pathways. In this sense, preparedness is not only about AI adoption itself, but also about the capacity to sustain productive employment creation and structural transformation in a changing technological landscape.

Digital connectivity introduces a further dimension of asymmetry. Workers in highly exposed occupations tend to be more digitally connected across all income levels, including in lower-income ASEAN economies, suggesting that displacement risks may materialize faster than often assumed, even where aggregate exposure is low. At the same time, an estimated 66.9 million workers globally in occupations with presumably high augmentation potential lack internet access, constraining the realization of

²¹ ASEAN-5 comprises Indonesia, Malaysia, the Philippines, Singapore and Thailand.

²² For example, the Stanford AI Index (2026) lists Singapore as having developed one “notable AI” in 2025, which compares with 59 for the United States of America and 35 for China. Similarly, Singapore registered 1.3 patents in 2024 per 100,000 inhabitants, which compares with 14.3 for the Republic of Korea and 12.1 for Luxembourg.

²³ See, in particular, the UNCTAD Frontier Technologies Readiness Index, which seems to indicate consistent financial gaps undermining the private sector's capacity to acquire, build, and deploy cutting-edge technologies.

potential productivity gains and increasing concerns around inequality.²⁴

Overall, ASEAN countries are entering the AI transition from markedly different starting points. As AI adoption

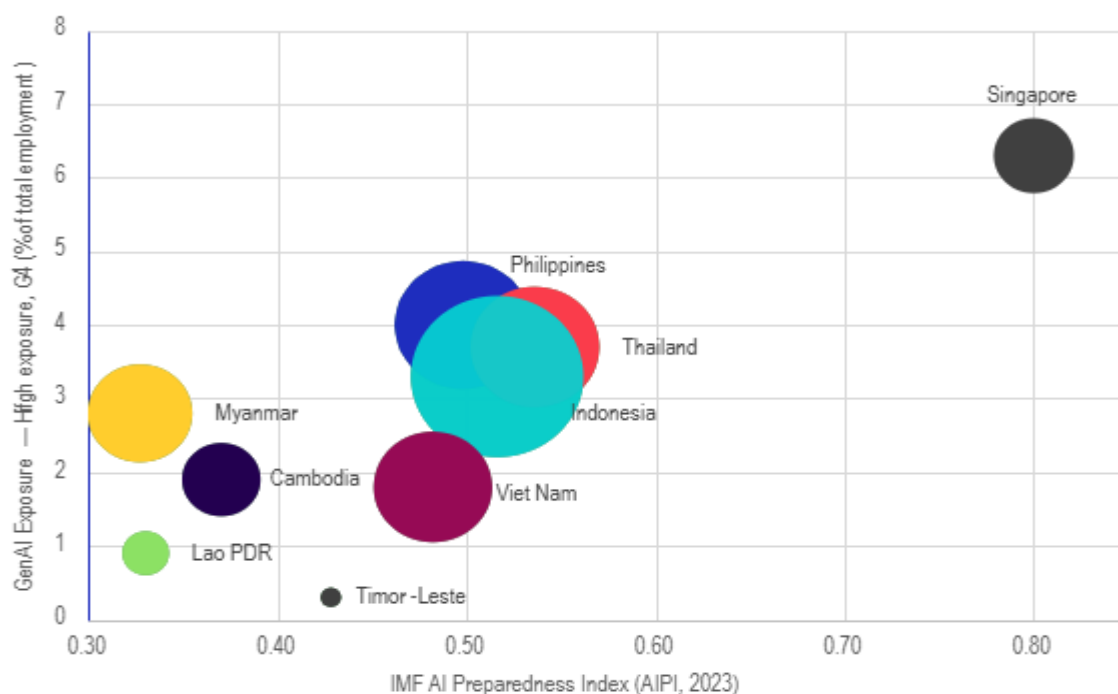
expands, differences in skills, infrastructure, institutional capacity and governance are likely to play a critical role in determining how the benefits and adjustment pressures associated with GenAI are distributed across workers, enterprises and economies.

► **Box 4. AI exposure and readiness do not always go hand in hand**

Labour market exposure and AI readiness do not always move together. Singapore stands out as the only economy combining both the highest preparedness and the highest exposure. However, several economies illustrate that similar levels of exposure can be associated with different levels of preparedness. For example, the Philippines ranks second on exposure but only fourth on preparedness, while Myanmar combines one of the lowest preparedness scores in the sample with a higher level of exposure than Cambodia, Viet Nam and Lao People's Democratic Republic. This suggests that a non-trivial share of workers may face potential labour market disruption in an environment with more limited institutional, digital and policy capacity to respond.

A positive association between preparedness and exposure is visible across the economies examined. However, the differences observed among countries with comparable exposure levels suggest that future labour market outcomes will depend not only on where workers are exposed to AI, but also on the capabilities available to support adaptation, innovation and productivity growth.

► **Figure B3. AI Preparedness Index and GenAI Exposure**



Note: Bubble size proportional to employment highly exposed to GenAI (G4). Source: IMF AI Preparedness Index (2023); ILO estimates based on ILO Microdata Repository and Gmyrek et al. 2025; for Singapore, the data point on Gen AI exposure is based on Gmyrek, Viollaz and Winkler 2026.

²⁴ See Gmyrek, Viollaz and Winkler 2026.

► Conclusions and regional priorities

The findings of this brief suggest that the potential for GenAI-driven labour market transformation in ASEAN is significant, but that widespread disruption is not yet visible. While around 80 million workers, corresponding to 22.9 per cent of total employment, are employed in occupations with more than minimal exposure to GenAI, only a relatively small share of 3.3 per cent are concentrated in occupations with the highest levels of exposure. Employment in highly exposed occupations has continued to expand, consistent with broader economic and structural trends; while AI adoption remains uneven and at a relatively early stage across much of the region.

At the same time, the absence of widespread disruption should not be interpreted as evidence that GenAI is having no effect on labour markets. Some indicators suggest that pressures may be emerging among younger workers and in selected entry-level occupations. However, these developments have not been accompanied by marked increases in youth unemployment or NEET rates, suggesting that recent changes may reflect shifts in hiring patterns, changed education choices and occupational reallocation rather than a broad deterioration of labour market outcomes for young people. Moreover, GenAI adoption remains at an early stage and labour market effects may take time to materialize.

Beyond employment effects, GenAI is also likely to reshape work organization, job design and the allocation of tasks within occupations. As AI adoption expands, labour market outcomes will depend not only on the number of jobs created or displaced, but also on how work is organized, the quality of jobs and the extent to which workers retain autonomy and opportunities for skills development. AI may also influence the intensity of work, the monitoring and management of workers, and the degree of control that workers exercise over their tasks and working conditions.²⁵

To ensure that AI innovation contributes to productivity growth, creates quality employment and fosters inclusive and equitable benefits across ASEAN, key priorities should be considered.

First, strengthening AI governance structures and policy integration through a human-centred approach is critical. It is key to actively involve ministries responsible for employment, labour and social protection in the building of the evidence base and in the design and implementation of AI-related policies, both at the national level and the regional level, and to build strong and gender-responsive social dialogue mechanisms that integrate workers' and employers' perspectives to inform policy decisions. Greater policy integration is also needed to address the complex and multi-faceted AI governance issues related to ethics, innovation, human resources development, and labour and social protections.

Second, in addition to suitable education policies to prepare tomorrow's workforce, inclusive labour market policies are needed to strengthen worker resilience and support labour market transitions. Expanding opportunities for AI-centred upskilling and reskilling, strengthening employment services and career development support, and ensuring the availability of timely and reliable labour market intelligence will help workers adapt to changing skills demands. Particular attention should be given to women, given their concentration in more exposed occupations (figure 2a) and the barriers they face in entering STEM education and related fields, which may increase the risk that GenAI exacerbates existing inequalities. Young people also warrant close attention, given emerging differences in employment outcomes observed among some young workers in highly exposed and entry-level occupations (figure 2b, box 3). For workers vulnerable to AI-driven labour market adjustments, ensuring income security can help facilitate transitions and enable retraining and re-entry into productive employment.

Third, supporting AI adoption and capability development among enterprises will be essential to ensure that productivity gains are widely shared and, in particular, not limited to the formal sector. As current evidence suggests that AI adoption remains concentrated in a relatively narrow group of firms (box 1), policies should focus on helping more enterprises – particularly micro, small and medium enterprises (MSMEs)

²⁵ See also Janine Berg and Hannah Johnston, "AI in Human Resource Management: The Limits of Empiricism", ILO Working Paper No. 154, November 2025.

– overcome barriers to adoption. This includes improving access to finance, strengthening knowledge diffusion from frontier firms, and supporting workplace-level capabilities to integrate AI technologies effectively and responsibly.

The extent to which productivity gains can be shared and translated into higher wages, better working conditions and broader social benefits will depend on institutions, social dialogue and the mechanisms through which these gains are shared between workers and enterprises.

Fourth, sustained and greater investment in digital infrastructure, skills and the broader innovation ecosystem will be critical for AI diffusion and inclusive development. Significant differences in AI readiness suggest that countries are entering the AI transition from markedly different starting points (box 4). Expanding reliable and affordable digital connectivity, strengthening digital and AI-related skills and fostering innovation capacity will be important to ensure that countries can both adopt AI and benefit from its productivity-enhancing potential.

Finally, stronger regional cooperation can help ASEAN countries manage a common technological transition while responding to diverse national circumstances.

Priorities include deepening cross-country knowledge exchange and sharing of innovative practices, as well as strengthening regional monitoring and surveillance of AI-related labour market trends, building on and complementing national data systems. In addition, advancing a more coordinated approach to human resources development will also be critical, by continuing to embed AI-related skills into regional ASEAN qualification and competency frameworks in a way that remains aligned with and adaptable to national systems. A strong regional approach can support greater policy coherence and help ensure the benefits of AI are shared in a more inclusive and balanced manner across all countries in ASEAN.

While GenAI has immense potential to transform the world of work, it must be approached as a tool to be mastered rather than a solution to all. Social dialogue mechanisms are essential for enabling dynamic adjustments in rapidly changing environments, helping to generate solutions that deliver more widely shared benefits. Ultimately, future labour market outcomes will depend less on exposure alone than on the policy choices to build the preparedness and resilience of workers, enterprises and institutions to adapt and navigate the AI transition.

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Contact details

ILO Regional Office for Asia and the Pacific
United Nations Building Bangkok
10200 Thailand

T: +66 2288 1234
E: BANGKOK@ilo.org
W: www.ilo.org/asiapacific